Yunian Pan

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Education

2021 – 2026 **Ph.D., NYU** in Electrical Engineering.
Committee: Prof. Quanyan Zhu, Prof. Elza Erkip & Prof. Sundeep Rangan
Thesis: *Variational Foundations of Multi-Agent Learning: A Unified Non-equilibrium Theory*

2018 – 2020 **M.Sc., NYU** in Electrical Engineering.
Thesis: *Urban vaccination site covering via semi-discrete Optimal Transport (OT).*

2014 – 2018 **BEng., BUPT** in Communication Engineering.

Working Experience

2019 – 2025 **Teaching & Graduate Assistant** — NYU Tandon, ECE Larx Lab
Taught and assisted graduate-level courses in system optimization, probability, robotics, and game theory.
Led research projects on large language model (LLM) multi-agent security, including jailbreaking analysis, agentic orchestration, fine-tuning, and red-teaming dataset construction.

2024 **Research Intern** — Augur Lab
Developed a signature-method-based market simulator for derivative instruments.
Applied general minimax-concave (GMC) regularized spline fitting for financial time-series modeling.

2025 **Research Intern** — NXP Semiconductors
Developed a high-fidelity multi-agent FMCW radar simulation framework with advanced signal filtering techniques.
Designed decentralized no-regret algorithms for FMCW radar interference mitigation.

Publications

Journal Articles

1. Y. Pan, T. Li, and Q. Zhu, "Model-agnostic hessian-free meta-policy optimization via zeroth-order estimation: A linear quadratic regulator perspective," *Dynamic Games and Applications*, 2025. DOI: 10.1007/s13235-025-00693-9.
2. Y. Pan and Q. Zhu, "A games-in-games paradigm for strategic hybrid jump-diffusions: Hamilton-jacobi-isaacs hierarchy and spectral structure," 2025, Under submission to IFAC *Nonlinear Analysis: Hybrid Systems*. arXiv: 2512.18098 [eess.SY]. URL: https://arxiv.org/abs/2512.18098.
3. Y. Pan, J. Li, L. Xu, S. Sun, and Q. Zhu, "Decentralized no-regret frequency-time scheduling for fmcw radar interference avoidance," 2025, Under submission to *IEEE Transactions on Aerospace and Electronic Systems*. arXiv: 2512.24619 [eess.SY]. URL: https://arxiv.org/abs/2512.24619.
4. T. Li, H. Li, Y. Pan, T. Xu, Z. Zheng, and Q. Zhu, "Meta stackelberg game: Robust federated learning against adaptive and mixed poisoning attacks," *arXiv preprint arXiv:2410.17431*, 2024, Under submission to *IEEE Transactions on Information Forensics and Security*.

Conference Proceedings and Preprints

1. Y. Pan, T. Li, and Q. Zhu, "On the resilience of traffic networks under non-equilibrium learning," in *2023 American Control Conference (ACC)*, IEEE, 2023, pp. 3484–3489.

2. Y. Pan, G. Peng, J. Chen, and Q. Zhu, "Masage: Model-agnostic sequential and adaptive game estimation," in *International Conference on Decision and Game Theory for Security*, Springer, 2020, pp. 365–384.
3. Y. Pan and Q. Zhu, "Efficient episodic learning of nonstationary and unknown zero-sum games using expert game ensembles," in *2021 60th IEEE Conference on Decision and Control (CDC)*, IEEE, 2021, pp. 1669–1676.
4. Y. Pan, T. Li, H. Li, T. Xu, Q. Zhu, and Z. Zheng, "A first order meta stackelberg method for robust federated learning," in *The Second Workshop on New Frontiers in Adversarial Machine Learning*, 2023. URL: `https://openreview.net/forum?id=sHH1KSlPcj`.
5. Y. Pan, T. Li, and Q. Zhu, "Is stochastic mirror descent vulnerable to adversarial delay attacks? a traffic assignment resilience study," in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 8328–8333. DOI: `10.1109/CDC49753.2023.10384003`.
6. Y. Pan and Q. Zhu, "On poisoned wardrop equilibrium in congestion games," in *International Conference on Decision and Game Theory for Security*, Springer, 2022, pp. 191–211.
7. Y. Pan, T. Li, and Q. Zhu, "On the variational interpretation of mirror play in monotone games," in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, 2024, pp. 6799–6804. DOI: `10.1109/CDC56724.2024.10885800`.
8. Y. Pan, T. Li, H. Li, T. Xu, Z. Q., and Z. Zheng, "A first order meta stackelberg method for robust federated learning," ICML Workshop on New Frontiers in Adversarial Machine Learning (AdvML-Frontiers'23). URL: `https://par.nsf.gov/biblio/10504770`.
9. Y.-T. Yang, Y. Pan, and Q. Zhu, "Preference-centric route recommendation: Equilibrium, learning, and provable efficiency," 2025. arXiv: `2504.01192 [cs.GT]`. URL: `https://arxiv.org/abs/2504.01192`.
10. Y. Pan, J. Li, L. Xu, S. Sun, and Q. Zhu, "A game-theoretic approach for high-resolution automotive fmcw radar interference avoidance," 2025, arXiv–2503.
11. Y. Pan and Q. Zhu, "Timing-aware two-player stochastic games with self-triggered control," 2025. arXiv: `2512.18109 [eess.SY]`. URL: `https://arxiv.org/abs/2512.18109`.
12. Y. Pan and Q. Zhu, "Bayesian holonic systems: Equilibrium, uniqueness, and computation," 2025. arXiv: `2512.18112 [eess.SY]`. URL: `https://arxiv.org/abs/2512.18112`.
13. Y. Pan and Q. Zhu, "Extending no-regret hopping in fmcw radar interference avoidance," 2, vol. 53, New York, NY, USA: Association for Computing Machinery, Aug. 2025, pp. 131–133. DOI: `10.1145/3764944.3764978`.

Book Chapters and Technical Reports

1. H. Li, T. Xu, T. Li, Y. Pan, Q. Zhu, and Z. Zheng, *A First Order Meta Stackelberg Method for Robust Federated Learning (Technical Report)*. 2023. arXiv: `2306.13273 [cs.CR]`.
2. T. Li, Y. Pan, and Q. Zhu, *Decision-dominant strategic defense against lateral movement for 5g zero-trust multi-domain networks*. Springer, 2024, pp. 25–76.
3. T. Li, Y.-T. Yang, Y. Pan, and Q. Zhu, *From Texts to Shields: Convergence of Large Language Models and Cybersecurity*. 2025.

Selected Projects

2025 **Game-Theoretic Market Making under Adversarial Risk.**
Developed a multi-layer game-theoretic extension of Avellaneda–Stoikov market making for adversarial environments, modeling strategic drift manipulation and inventory-dependent exploitation. Built a high-fidelity simulation platform incorporating stochastic volatility and regime-switching dynamics, calibrated on cryptocurrency market data.

Deep Hedging with Reinforcement Learning under Partial Observability.
Designed an RL-based hedging framework for multi-asset options in partially observable markets, using approximate information states to handle latent volatility. Simulated high-dimensional price paths via multi-factor stochastic volatility models. Implemented RNN/Transformer deep hedging agents to achieve PnL-efficient hedging.

2024–2025 **Decentralized Learning for FMCW Radar Interference Mitigation.**
Developed a multi-agent radar simulation framework modeling mutual interference in dense FMCW sensing environments. Designed decentralized no-regret learning algorithms enabling autonomous radars to adapt transmission strategies without coordination. Demonstrated convergence to stable interference-aware operating points and significant performance gains in detection and estimation accuracy under realistic signal and noise models.

Technical Skills

Mathematical Foundations	Statistical Learning • Convex Optimization • Algorithmic Game Theory • Probability & Stochastic Calculus • Information Theory • Signal Processing • Optimal Control • Reinforcement Learning
Tools & Platforms	NumPy • PyTorch • Gym • Pettingzoo • SUMO • R • MATLAB • PostgreSQL • LoRA • vLLM • VectorBT • QuantLib • Gcloud • AWS Bedrock
Supporting Capabilities	Academic Research & Presentations • Linux • Git • LaTeX

Miscellaneous

Awards and Achievements

2023 **Dante Youla Award for Graduate Research Excellence**, NYU Tandon.

2022 **Best Paper Award**, GameSec Conference 2022.

2020 **Merit Award**, Outstanding academic performance in ECE Department.

Academic Service

2025 **Technical Session Chair**, IEEE International Radar Conference 2025.

Session Co-Organizer, *Digital twin-based Driver risk-aware intelligent mobility analytics for urban transportation management*

2024–2025 **Peer Reviewer**, Automatica, Annals of Reviews in Control, IEEE CDC, ACC, INFOCOM, CNS.

Invited Conference Sessions

2025 **INFORMS Annual Meeting**, *Computational Foundations of Multi-agent learning* Atlanta, GA.

IEEE CDC, *Bounded Rationality in Human-AI Decision-Making* (Rio de Janeiro, Brazil).

IEEE International Radar, *Automotive Radar Principles and Challenges* Atlanta, GA.

2024 **IEEE CDC**, *Networks, Games and Learning II* (Milan, Italy).

SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, GA.

2023 **IEEE ACC**, *Resiliency and Privacy Throughout Networked Cyber-Physical Systems* (San Diego, CA).